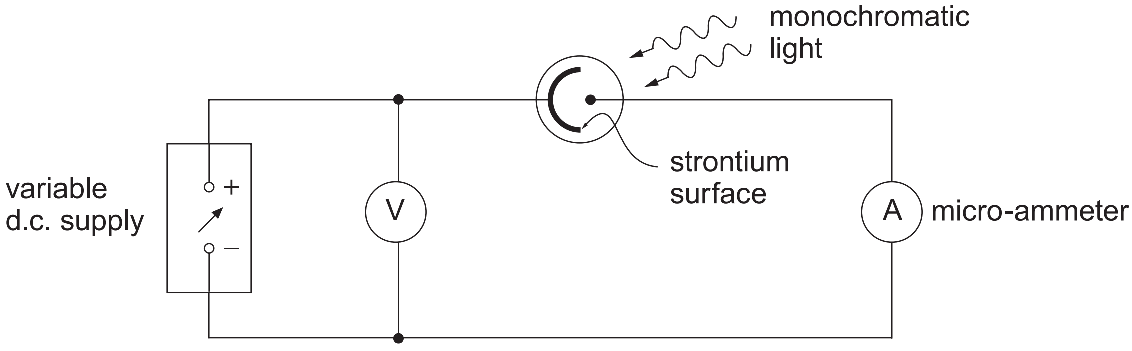
Examiner only

7. *(a)* A student uses the apparatus shown in the diagram to determine the maximum kinetic energy, $E_{k\,max}$, of electrons ejected from a strontium surface by light of wavelength 405 nm.



He records that $E_{k\,max}$ = 0.480 eV.

(i) Describe briefly how he used the apparatus to obtain this result. [2]

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(ii) Determine a value for the work function of strontium from this result. [3]

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Examiner only

(iii) The student next attempts to measure $E_{k\,max}$ for the same surface, but with light of wavelength 650 nm. He is surprised to find that the micro-ammeter reads zero for **all** applied pds – even when he makes the light brighter. Explain in terms of photons why he should not be surprised, justifying your answer with a calculation. [3]

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(b) (i) Calculate the speed at which an electron must be moving to have a de Broglie wavelength of 5.0×10^{-11} m. [2]

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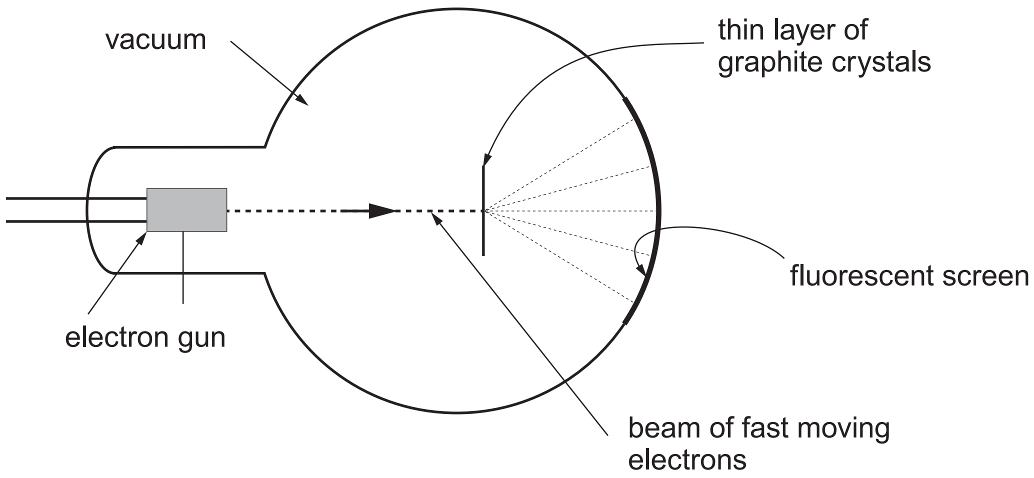
Examiner only

(ii) The diagram shows one way in which an aspect of the nature of electrons may be demonstrated by experiment.



Several bright concentric circles, caused by the impact of electrons on the fluorescent screen are seen. Explain briefly how the pattern of circles arises. [2]

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Turn over.